A RATIONAL COUNTEREXAMPLE TO THE RECIPROCAL-LOGARITHMIC ASSERTION IN ERDŐS PROBLEM 143

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ABSTRACT. We construct a countably infinite set $A \subset \mathbb{Q} \cap [2, \infty)$ such that $|ka - b| \geq 1$ for all distinct $a, b \in A$ and all positive integers k, while $\sum_{a \in A} 1/(a \log a) = \infty$. This gives a negative answer to the reciprocal-logarithmic convergence question in Erdős problem 143, distinct from its logarithmic-density question. The construction combines a primitive-set selection lemma for periodic sets with rational blocks equipped with private primes. At the beginning of each round, a new integer grid makes the preceding configuration exclude only a small proportion of labels. Finitely many new blocks then add a uniformly positive amount of weight without moving earlier points. The selection lemma follows from Euler-product estimates, Dirichlet characters and Abel averaging at a fixed modulus. Every constructed point lies within $m^{-12}/2$ to the right of an integer m. A companion Lean 4 formalization checks the analytic lemma, the finite extensions and the complete existence theorem.

1. INTRODUCTION

Write $\mathbb{N} = \{1, 2, \ldots\}$ and let log denote the natural logarithm. For $x > 1$ and for a finite or countable set $B \subset (1, \infty)$, put

$$\mathrm{w}(x) = \frac{1}{x \log x}, \qquad W(B) = \sum_{x \in B} \mathrm{w}(x) \in [0, \infty].$$

The sum is understood as the supremum of the sums over finite subsets.

Definition 1.1. A set $A \subset (1, \infty)$ is *dilation-separated* if

$$|ka - b| \geq 1 \quad \text{for all } a, b \in A,\ a \neq b, \quad \text{and all } k \in \mathbb{N}. \tag{1.1}$$

The quantification is over ordered pairs, so both directions are required. A set $P \subset \mathbb{N} \setminus \{1\}$ is *primitive* if no element of P divides a distinct element of P.

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Taking $k = 1$ in (1.1) shows that distinct points are at least one apart. Hence every bounded interval contains only finitely many points of a dilation-separated set, and every such set is at most countable.

For a set of integers greater than one, condition (1.1) is equivalent to primitivity: an integer distance less than one must be zero, so the excluded event is precisely divisibility between distinct elements. Erdős proved that every primitive set of integers has finite reciprocal-logarithmic weight [2]. The analogous question for real sets is subtler, since exact divisibility is replaced by uniform separation from integer dilates.

In problem II on the final page of [3], Erdős asks whether condition (1.1) forces both convergence of $W(A)$ and vanishing logarithmic density. These are the two questions in the problem now numbered 143 in [1]. Koukoulopoulos, Lamzouri, and Lichtman [4, Theorem 1] established the latter assertion: every dilation-separated set satisfies

$$\sum_{a\in A\cap[1,x]} \frac{1}{a} = o(\log x). \tag{1.2}$$

Their theorem gives approximate integer-dilation relations under positive upper logarithmic density. The theorem below shows that divergence of $W(A)$ alone does not force such relations, even for rational sets whose points lie extremely close to integers. In particular, the two proposed sparsity conclusions have different answers; the counterexample is consistent with (1.2).

Theorem 1.2. *There exists a countably infinite set $A \subset \mathbb{Q} \cap [2, \infty)$ with no finite accumulation point such that* (1.1) *holds and*

$$W(A) = \infty. \tag{1.3}$$

Every $a \in A$ can be written as

$$a = m + \eta, \qquad m \in \mathbb{N}, \quad 0 < \eta \le \frac{1}{2} m^{-12}. \tag{1.4}$$

More precisely, the construction produces one nested sequence of finite dilation-separated sets $F_1 \subset F_2 \subset \cdots$ whose union is A and for which

$$W(F_R) \ge \frac{R}{576} \qquad (R \in \mathbb{N}). \tag{1.5}$$

The main difficulty is to accumulate infinite weight while maintaining separation from every integer multiple of every earlier point. A small perturbation of integer primitive sets alone does not give this global compatibility. We use finite rational blocks with a private-prime invariant. At the beginning of each round this invariant supplies a new grid on which all earlier points exclude only a small proportion of labels. The grid changes, but the earlier points do not. Periodic primitive-set selection then supplies enough new labels to obtain a uniform gain in weight. The constants are sufficient for this construction; no optimality is asserted for the gain or the exponent in (1.4).

The following sections give the argument in full. The companion formal proof has no additional analytic or construction hypothesis; its scope and reproduction instructions appear in Section 8.3.

1.1. **Outline and notation.** For $n \in \mathbb{N}$, let $\Omega(n)$ be the number of prime factors of n, counted with multiplicity; in particular, $\Omega(1) = 0$. Every subset of

$$\mathcal{L}_r = \{n \in \mathbb{N} : \Omega(n) = r\}, \qquad r \geq 1,$$

is primitive. Indeed, $n \mid n'$ with $n \neq n'$ implies $\Omega(n') = \Omega(n) + \Omega(n'/n) > \Omega(n)$.

A set $E \subset \mathbb{N}$ is *periodic of period* L if membership depends only on the residue modulo L, where $L \in \mathbb{N}$. Its density is

$$\mathrm{d}(E) = \frac{\#\{b \in \mathbb{Z}/L\mathbb{Z} : b \text{ is admitted by } E\}}{L}.$$

This definition is independent of the chosen period. All density calculations below take place on a finite common period. In particular, the union bound for finitely many periodic sets is simply the union bound for finite subsets of a residue ring.

The analytic part supplies a finite primitive subset of weight greater than $1/6$ in every periodic set of density greater than $2/3$, above any fixed threshold. The geometric part uses it to construct a block of weight greater than $1/(24D)$ while spending at most $3/D$ of the density of labels on a fixed grid $D\mathbb{N}$. A round contains $\lfloor D/12 \rfloor$ blocks. Private primes make it possible to restart the next round on a new grid while preserving the entire preceding configuration.

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Mathematical software and supporting material. The companion project uses Lean 4.24.0 and a pinned revision of mathlib. The predicate named `Original` in the code is precisely the dilation-separation condition (1.1). Its source files, exact theorem statements and reproduction instructions are supplied as supplementary material; the repository is `https://github.com/mapice/erdos-143-candidate-disproof`. The scope of the formal checks is described in Section 8.3.

2. Primitive weight in a periodic set

The classical analytic inputs in this section are the absolutely convergent Euler products for $\zeta(s)$ and Dirichlet L-functions for $\Re s > 1$, character orthogonality for a fixed modulus, and continuity of $L(s,\chi)$ at $s = 1$ for nonprincipal χ; see [5, Sections 2 and 6]. Non-vanishing at $s = 1$ is not needed. We use an Abel average to find arbitrarily high levels with sufficient weight, without requiring a lower bound for every sufficiently high individual level.

Theorem 2.1 (Periodic selection)**.** *Let $E \subset \mathbb{N}$ be periodic, with $\mathrm{d}(E) > 2/3$. For every integer $N \geq 2$, there is a finite primitive set $P \subset E \cap [N, \infty)$ such that $W(P) > 1/6$.*

For a fixed modulus $L \geq 1$ and residue $b \pmod L$, define

$$A_{L,b}(u) = \sum_{\substack{n \geq 2 \\ n \equiv b \pmod L}} u^{\Omega(n)} \mathrm{w}(n), \qquad 0 \leq u < 1. \tag{2.1}$$

We will prove convergence of this series and the sufficient estimate

$$\liminf_{u \uparrow 1} (1-u) A_{L,b}(u) \geq \frac{1}{4L}. \tag{2.2}$$

The modulus and residue are fixed before the limit is taken; no uniformity in L is required. This estimate will supply arbitrarily high primitive levels whose weight exceeds $1/6$ after summation over the admitted residue classes.

2.1. Uniform Euler-factor estimates.

For a Dirichlet character χ modulo L, $0 \leq u \leq 1$, and $t > 0$, put

$$G_\chi(u,t) = \prod_p (1 - u\chi(p)p^{-1-t})^{-1} = \sum_{n \geq 1} \frac{u^{\Omega(n)}\chi(n)}{n^{1+t}}. \tag{2.3}$$

Products indexed by p are over primes. Absolute convergence follows from $|\chi(n)| \leq 1$ and $u^{\Omega(n)} \leq 1$, or directly from finite Euler products and the summability of $\sum_p p^{-1-t}$. The $n = 1$ term is 1, including when $u = 0$.

Lemma 2.2. *For $0 \leq u \leq 1$ and $t > 0$,*

$$|G_\chi(u,t)|^2 \leq 2|L(1+t,\chi)|^{2u}. \tag{2.4}$$

For the principal character χ_0 modulo L, the product is real and positive and satisfies

$$G_{\chi_0}(u,t) \geq \big(L(1+t,\chi_0)/2\big)^u. \tag{2.5}$$

Proof. For $z \in \mathbb{C}$ with $|z| \leq 1/2$, set $A = |1-z|^2$ and $S = (1-u) + uA$. The identity

$$|1 - uz|^2 = S - u(1-u)|z|^2$$

and the bounds $S \geq 1/4$ and $u(1-u) \leq 1/4$ give

$$|1 - uz|^2 \geq S(1 - |z|^2) \geq A^u(1 - |z|^2),$$

where the last step is weighted arithmetic–geometric mean. Inverting,

$$|(1-uz)^{-1}|^2 \le |(1-z)^{-1}|^{2u}(1-|z|^2)^{-1}. \tag{2.6}$$

The quadratic corrections have a uniform bound. For any finite set T of primes and $M \ge \max(T \cup \{2\})$,

$$\prod_{p\in T}(1-p^{-2})^{-1} \le \prod_{n=2}^{M}(1-n^{-2})^{-1} = \frac{2M}{M+1} < 2. \tag{2.7}$$

Apply (2.6) with $z = \chi(p)p^{-1-t}$ and pass to the convergent Euler products to obtain (2.4).

For $0 \le x < 1$, concavity and an elementary rearrangement give

$$(1+x)^u \le 1+ux \le (1-ux)^{-1}.$$

For the principal character use $x = p^{-1-t}$ when $p \nmid L$ and $x = 0$ otherwise. Since $1+x = (1-x)^{-1}(1-x^2)$, (2.7) implies

$$G_{\chi_0}(u,t) \ge \left(\prod_{p\nmid L}(1+p^{-1-t})\right)^u \ge \big(L(1+t,\chi_0)/2\big)^u.$$

The same argument applies to finite products first; convergence then justifies the displayed passage to infinite products. □

Write G_1 for the unrestricted product, corresponding to modulus one, and put

$$\kappa_L = \prod_{p\mid L}(1-p^{-1}) = \frac{\varphi(L)}{L}, \qquad \varphi(1) = 1.$$

Lemma 2.3. *For a fixed nonprincipal character χ,*

$$|G_\chi(u,t)| \le C_{L,\chi} \qquad (0 \le u \le 1,\ 0 < t \le 1). \tag{2.8}$$

There is $\varepsilon \in (0,1]$, depending on L, such that

$$G_{\chi_0}(u,t) \ge \left(\frac{\kappa_L}{4t}\right)^u \qquad (0 \le u \le 1,\ 0 < t \le \varepsilon). \tag{2.9}$$

The unrestricted product satisfies

$$G_1(u,t) \le 4t^{-u} \qquad (0 \le u \le 1,\ 0 < t \le 1). \tag{2.10}$$

Uniformly in all characters and $0 \le u \le 1$,

$$|G_\chi(u,t) - 1| \le C2^{-t} \qquad (t \ge 1), \tag{2.11}$$

with an absolute constant C.

Proof. For nonprincipal χ, continuity on the compact real segment $[1,2]$ gives a bound $B \ge 1$ with $|L(s,\chi)| \le B$ there. For $0 < u \le 1$, equation (2.4) gives $|G_\chi(u,t)|^2 \le 2B^{2u} \le 2B^2$, while $G_\chi(0,t) = 1$. This proves (2.8) with $C_{L,\chi} = 2B$. The principal Euler product and the elementary estimates

$$\frac{1}{t} \le \zeta(1+t) \le 1 + \frac{1}{t}$$

give $tL(1+t,\chi_0)\to\kappa_L$ as $t\downarrow 0$. In particular, $L(1+t,\chi_0)\ge\kappa_L/(2t)$ for small t. Equation (2.5) proves (2.9). For modulus one, (2.4) gives, when $0<t\le 1$,

$$G_1(u,t)\le\sqrt{2}\,\zeta(1+t)^u\le\sqrt{2}(2/t)^u\le 4t^{-u}.$$

Finally, by the absolutely convergent series in (2.3),

$$|G_\chi(u,t)-1|\le\sum_{n\ge 2}n^{-1-t}\le 2^{-(t-1)}\sum_{n\ge 2}n^{-2}\qquad(t\ge 1).$$

This proves (2.11). □

2.2. **Integration and the character averages.**

Lemma 2.4. *For $0\le u<1$, the series $\sum_{n\ge 2}u^{\Omega(n)}\mathrm{w}(n)$ converges. For every character χ,*

$$J_\chi(u):=\int_0^\infty(G_\chi(u,t)-1)\,dt=\sum_{n\ge 2}\chi(n)u^{\Omega(n)}\mathrm{w}(n),\tag{2.12}$$

with an absolutely convergent integral and series. Moreover,

$$C_2:=\sum_{n\ge 2}\frac{1}{n(\log n)^2}<\infty.\tag{2.13}$$

Proof. The unrestricted nonnegative integrand $G_1(u,t)-1$ is integrable: near zero use (2.10) and $u<1$, and near infinity use (2.11). Tonelli's theorem and the elementary identity

$$\int_0^\infty n^{-1-t}\,dt=\frac{1}{n\log n}\qquad(n\ge 2)$$

therefore prove convergence of the unrestricted weighted series and (2.12) for the trivial character. For a general character, the sum of the integrals of the absolute values is bounded by this same convergent series. Termwise integration is consequently justified and proves (2.12).

For the last assertion, $G_1(1,t)=\zeta(1+t)$. Thus $t(\zeta(1+t)-1)$ is bounded near zero by the elementary zeta estimates, and decays at infinity by (2.11). A further application of Tonelli, now using

$$\int_0^\infty tn^{-1-t}\,dt=\frac{1}{n(\log n)^2},$$

proves (2.13). □

Lemma 2.5. *For each nonprincipal χ,*

$$(1-u)J_\chi(u)\longrightarrow 0\qquad(u\uparrow 1).\tag{2.14}$$

For the principal character,

$$\liminf_{u\uparrow 1}(1-u)J_{\chi_0}(u)\ge\frac{\kappa_L}{4}.\tag{2.15}$$

Proof. For nonprincipal χ, equations (2.8) and (2.11) provide an integrable majorant independent of $0 \le u < 1$ for $|G_\chi(u,t) - 1|$. Thus $J_\chi(u)$ stays bounded, which proves (2.14). For χ_0, every term of $G_{\chi_0}(u,t) - 1$ is nonnegative. Integrating the lower bound (2.9) on $(0, \varepsilon)$ gives

$$J_{\chi_0}(u) \ge \frac{(\kappa_L/4)^u \varepsilon^{1-u}}{1-u} - \varepsilon.$$

After multiplication by $1-u$, the right side tends to $\kappa_L/4$. □

2.3. **Every residue class.**

Proposition 2.6. *Estimate* (2.2) *holds for every fixed* $L \ge 1$ *and every residue* $b \pmod L$*, including noninvertible residues.*

Proof. For an invertible residue, character orthogonality and the absolute convergence in Lemma 2.4 give

$$A_{L,b}(u) = \frac{1}{\varphi(L)} \sum_{\chi \bmod L} \overline{\chi(b)} J_\chi(u). \tag{2.16}$$

The principal term has normalized lower limit at least $\kappa_L/(4\varphi(L)) = 1/(4L)$. The nonprincipal terms, after multiplication by $1-u$, tend to zero by Lemma 2.5. Taking real parts in (2.16) proves the assertion. For $L = 1$, the same argument has only the principal term.

For a general residue use its representative $b \in \{0, \dots, L-1\}$ and put $g = \gcd(b, L)$, $M = L/g$, and $c = b/g$, with $\gcd(0, L) = L$. The residue $c \pmod M$ is invertible, also when $M = 1$ with its unique class. Multiplication $m \mapsto gm$ sends the integers $m \ge 2$ in that class injectively into the integers $n \ge 2$ in the original class, and $\Omega(gm) = \Omega(g) + \Omega(m)$. For $m \ge 2$,

$$0 \le \frac{1}{m \log m} - \frac{1}{m(\log m + \log g)} \le \frac{\log g}{m(\log m)^2}. \tag{2.17}$$

Using $0 \le u^{\Omega(m)} \le 1$, $u^{\Omega(g)} \le 1$, and Lemma 2.4, we get

$$A_{L,b}(u) \ge \frac{u^{\Omega(g)}}{g} A_{M,c}(u) - \frac{\log g}{g} C_2. \tag{2.18}$$

An omitted target term corresponding to $m = 1$, if present, has $n = g \ge 2$ and is nonnegative, so does not affect the lower bound. Multiplication by $1-u$ makes the error term vanish. The first coefficient tends to $1/g > 0$, and the coprime case supplies the lower bound $1/(4M) > 0$. The definition of lower limit now gives $1/(4gM) = 1/(4L)$. No coprimality between g and m is required. □

2.4. **Heavy levels and finite extraction.**

Proof of Theorem 2.1. Fix a period L and its set S of admitted residue classes. Sum Proposition 2.6 over this finite set to obtain

$$\liminf_{u\uparrow 1}(1-u)\sum_{\substack{n\geq 2\\ n\in E}} u^{\Omega(n)}\mathrm{w}(n) \geq \frac{|S|}{4L} = \frac{\mathrm{d}(E)}{4} > \frac{1}{6}. \tag{2.19}$$

For $r \geq 0$, put

$$f_r = \sum_{\substack{n\geq 2\\ n\in E\\ \Omega(n)=r}} \mathrm{w}(n).$$

Each f_r is finite: the convergent weighted series at $u = 1/2$ bounds it after multiplication by 2^r. Regrouping the nonnegative summable series gives, for $0 \leq u < 1$,

$$\sum_{\substack{n\geq 2\\ n\in E}} u^{\Omega(n)}\mathrm{w}(n) = \sum_{r\geq 0} u^r f_r. \tag{2.20}$$

If $f_r \leq 1/6$ for all $r \geq R$, then

$$(1-u)\sum_{r\geq 0} u^r f_r \leq \frac{1}{6} + (1-u)\sum_{r<R} f_r.$$

Its upper limit would be at most $1/6$, contradicting (2.19). Thus there are arbitrarily high r with $f_r > 1/6$. Choose such an $r \geq N \geq 2$. Every integer on that level is at least $2^r \geq r \geq N$, and the level is primitive. Since its weight is a convergent positive series exceeding $1/6$, a finite truncation still exceeds $1/6$. This truncation is the required set P. □

3. Finite configurations and private primes

We index blocks by $i = 1, 2, \ldots$ in their order of insertion. A block has data

$$\begin{gathered} D_i \in \mathbb{N}, \quad D_i \geq 48, \quad P_i \subset \mathbb{N}\setminus\{1\} \text{ finite, nonempty, primitive,} \\ H_i, G_i \text{ primes}, \qquad p_i = H_i G_i, \quad q_i = p_i - 1, \quad c_i = \frac{p_i}{q_i}, \\ A_i = \{a_{i,n} : n \in P_i\}, \qquad a_{i,n} = c_i D_i n = \frac{H_i N_{i,n}}{q_i}, \quad N_{i,n} = G_i D_i n. \end{gathered} \tag{3.1}$$

The prime H_i is called the *private prime* of the block. Since $q_i \equiv -1 \pmod{H_i}$, the factor H_i in the numerator cannot cancel against the denominator. The invariant below prevents this prime from dividing the nonprivate numerator parts of any block.

Definition 3.1. A finite list of blocks is an *admissible configuration* if, writing $F = \bigcup_{i=1}^{t} A_i$, the following hold:

(I1) F is dilation-separated, and every point of A_i exceeds every point of $\bigcup_{j<i} A_j$ by more than 1.

(I2) The primes H_i are distinct and greater than 2, and

$$H_j \nmid N_{i,n} \qquad (1 \le i, j \le t,\ n \in P_i). \tag{3.2}$$

(I3) Every point $a_{i,n}$ satisfies $a_{i,n}/H_i > 4$ and (1.4) with $m = D_i n$.

(I4) The private primes satisfy the strict finite budget

$$\sum_{i=1}^{t} \frac{1}{H_i} < \frac{1}{100}. \tag{3.3}$$

The empty list is admissible, with $F = \varnothing$. We write $M(F) = \max F$ when $F \neq \varnothing$, and $M(\varnothing) = 0$.

All invariants concern only data that have already been selected. A new private prime is chosen large enough to preserve the remaining positive budget; no prescribed rate of prime growth is needed.

4. A grid compatible with the preceding configuration

Lemma 4.1. *Given an admissible configuration with point set F_0, there exists an integer $D \ge 48$ such that*

$$\begin{aligned} N_{i,n} \mid D \quad &\text{for all preceding points,} \\ (D, H_i) = 1 \quad &\text{for all preceding private primes.} \end{aligned} \tag{4.1}$$

For this D, if $a = a_{i,n} \in F_0$, $\ell \in \mathbb{N}$, and $H_i \nmid \ell$, then

$$|ka - D\ell| > 4 \qquad \text{for every } k \in \mathbb{Z}. \tag{4.2}$$

If instead $H_i \mid \ell$, then $D\ell$ is a positive integer multiple of a.

Proof. For $F_0 = \varnothing$, take $D = 48$. Otherwise let Q be the product of the distinct positive integers $N_{i,n}$ stored in F_0. By (3.2), Q is coprime to every preceding private prime. All these primes are odd. Thus $D = 2^v Q$, with $v \ge 0$ sufficiently large, satisfies (4.1).

Fix a preceding point $a = H_i N_{i,n}/q_i$ and put $t_a = D/N_{i,n}$. This is a positive integer. Since $H_i \nmid D$ and $q_i \equiv -1 \pmod{H_i}$, we have $(t_a q_i, H_i) = 1$. The exact identity

$$|ka - D\ell| = \frac{a}{H_i}|kH_i - t_a q_i \ell| \tag{4.3}$$

holds for every $k \in \mathbb{Z}$. If $H_i \nmid \ell$, its integer factor is nonzero, so (4.2) follows from (I3). If $H_i \mid \ell$, then $D\ell/a = t_a q_i \ell/H_i$ is a positive integer. The proof uses the displayed representation of a and does not require that representation to be reduced. □

For a finite rational set $F \subset (0, \infty)$ and an integer $D \ge 1$, define the set of admissible labels

$$\mathcal{R}(F, D) = \{\ell \in \mathbb{N} : ka - D\ell \notin (-1, 1] \text{ for every } a \in F \text{ and every } k \in \mathbb{Z}\}. \tag{4.4}$$

The half-open interval is chosen for the later positive perturbation.

Lemma 4.2. *The set $\mathcal{R}(F, D)$ is periodic. For $F = F_0$ and D as in Lemma 4.1,*

$$\mathcal{R}(F_0, D) = \{\ell \in \mathbb{N} : H_i \nmid \ell \text{ for every preceding } H_i\}, \tag{4.5}$$

and in particular

$$\mathrm{d}(\mathcal{R}(F_0, D)) > \frac{99}{100}. \tag{4.6}$$

Proof. Write each $a \in F$ as A_a/B_a with positive integer numerator and denominator. Replacing ℓ by $\ell + A_a$ and k by $k + DB_a$ leaves $ka - D\ell$ unchanged. Thus the condition for this point is periodic with period A_a, and the product of these finitely many numerators is a common period. For $F = \varnothing$, every label is admitted and one may take period 1.

Equation (4.5) follows from the two alternatives in Lemma 4.1. By the Chinese remainder theorem its density is $\prod_i (1 - H_i^{-1})$, with value 1 for an empty product. The inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ for $0 \leq x_i \leq 1$, together with (3.3), proves (4.6). □

The reset in Lemma 4.1 changes only the grid used for future blocks. Every old source excludes just one class of labels on this grid, even though all of its integer multiples must be avoided.

5. The density cost of a block

For integers $D \geq 1$ and $p \geq 2D$, put $q = p - 1$, $c = p/q$, and $r = \lfloor q/D \rfloor$. Define a finite forbidden set of residues by

$$\mathcal{B}(p, D) = \{z \bmod p : z \in \mathbb{Z},\ -r \leq z \leq r\}. \tag{5.1}$$

The interval is closed. This slightly stronger exclusion gives the half-open admissibility required in (4.4).

Lemma 5.1 (Root-residue bound)**.** *Let $\ell \in \mathbb{N}$. If $\ell \bmod p \notin \mathcal{B}(p, D)$, then $|cDh - D\ell| > 1$ for every $h \in \mathbb{Z}$. Moreover,*

$$\frac{|\mathcal{B}(p, D)|}{p} \leq \frac{3}{D}. \tag{5.2}$$

Consequently the same exclusion separates $D\ell$ from every integer multiple of every point in any block $\{cDn : n \in P\}$.

Proof. If $|cDh - D\ell| \leq 1$, then $D|ph - q\ell| \leq q$. The integer $z = ph - q\ell$ therefore lies in $[-r, r]$. Since $q = p - 1$, we have $z \equiv \ell \pmod{p}$, proving the first assertion by contraposition. Reduction modulo p cannot increase cardinality, so $|\mathcal{B}(p, D)| \leq 2r + 1$. Hence

$$\frac{|\mathcal{B}(p, D)|}{p} \leq \frac{2q}{pD} + \frac{1}{p} \leq \frac{2}{D} + \frac{1}{D} = \frac{3}{D}.$$

The last estimate already holds for $p \geq D$. For a point cDn, its k-fold multiple is $cD(kn)$, so the first assertion applies with $h = kn$. No coprimality between D and p is assumed. □

Remark 5.2. The number cD is a counting device only. It is not added to the separated set. Its multiples contain all multiples of the entire block, which is why the bound is independent of the number of labels in P.

Fix a grid D for the configuration F_0 at the start of a round, and put

$$J = \left\lfloor \frac{D}{12} \right\rfloor. \tag{5.3}$$

After $s \le J$ insertions, let $\mathcal{P}$ be the set of the root numerators p used in this round, so $|\mathcal{P}| \le s$ and $p \ge 2D$ for $p \in \mathcal{P}$. Let $\mathcal{H}$ contain all current private primes, and possibly a newly chosen one, subject to $\sum_{H\in\mathcal{H}} H^{-1} < 1/100$. Set

$$E(\mathcal{H},\mathcal{P},D) = \left\{ \ell \in \mathbb{N} : \begin{array}{ll} H \nmid \ell & (H \in \mathcal{H}), \\ \ell \bmod p \notin \mathcal{B}(p,D) & (p \in \mathcal{P}) \end{array} \right\}. \tag{5.4}$$

This filter lies in $\mathcal{R}(F,D)$ for the entire current point set F: old-round points are protected by Lemma 4.1, and points inserted in this round by Lemma 5.1.

A common period is the product of the finitely many moduli in $\mathcal{H}$ and $\mathcal{P}$, with empty product 1. For each modulus m dividing that period, every residue modulo m has the same number of preimages. Thus lifting a forbidden set to the common period preserves its density. The finite union bound gives

$$\begin{aligned} \mathrm{d}(E) &\ge 1 - \sum_{H\in\mathcal{H}} \frac{1}{H} - \sum_{p\in\mathcal{P}} \frac{|\mathcal{B}(p,D)|}{p} \\ &> \frac{99}{100} - \frac{3s}{D} \ge \frac{37}{50} > \frac{2}{3}. \end{aligned} \tag{5.5}$$

No independence or coprimality of these moduli is required. The single private-prime budget covers both the old-round points and all private prime exclusions; it is not counted twice.

6. The next block

Retain the round setup of Section 5. The grid D was chosen for the initial configuration F_0 by Lemma 4.1; the current point set F consists of F_0 and the $s < J$ blocks already inserted on this same grid. Assume that the current configuration is admissible and that D is coprime to all its private primes. This holds at the start of the round by (4.1). We show that the next insertion exists and preserves these conditions; its new private prime will exceed D.

6.1. **Choosing the prime before the labels.** Let i be the index of the next block and let $\delta = 1/100 - \sum_{j<i} H_j^{-1} > 0$. Choose a prime H_i exceeding

$$2, \quad D, \quad \delta^{-1}, \quad \text{all preceding } H_j, \quad \text{all preceding integers } N_{j,n}. \tag{6.1}$$

Unboundedness of the primes meets these finitely many bounds. Since $H_i^{-1} < \delta$, the enlarged private-prime set still satisfies (3.3). Let $\mathcal{H} = \{H_1, \dots, H_i\}$, let $\mathcal{P}$ be the set of numerators p_j of the s blocks already inserted in this round, and define

$$E = E(\mathcal{H},\mathcal{P},D). \tag{6.2}$$

The entire periodic filter, and a common period, are now fixed. By (5.5),

$$\mathrm{d}(E) > \frac{37}{50} > \frac{2}{3}. \tag{6.3}$$

Also $E \subset \mathcal{R}(F, D)$ and every label in E avoids every current private prime, including H_i.

Apply Theorem 2.1 to this fixed E, with an integer threshold large enough to obtain a finite primitive $P_i \subset E$ with

$$\begin{aligned} W(P_i) > \frac{1}{6}, \qquad & \min P_i \geq 2D, \\ D \min P_i > 4H_i, \qquad & D \min P_i > M(F) + 1. \end{aligned} \tag{6.4}$$

Such a threshold exists because F, D, H_i are already fixed. The set P_i is nonempty, and we put $U = D \max P_i$.

6.2. **Choosing the perturbation after the labels.** For each $a \in F$, retain the reduced representation $a = A_a/B_a$ with positive integer denominator B_a. Choose a prime G_i, distinct from all $H_1, \ldots, H_i$, sufficiently large that, with $p = H_i G_i$, $q = p - 1$, and $c = p/q$,

$$\begin{aligned} p \geq 2D, \qquad 1 < c < 2, \qquad q \geq 2U^{13}, \\ \frac{U}{q} < \frac{1}{B_a} \qquad (a \in F). \end{aligned} \tag{6.5}$$

The last family is empty when $F = \varnothing$. All these conditions are finitely many lower bounds on G_i, together with finitely many prime exclusions, and so can be met simultaneously.

Set $D_i = D$ and define A_i by (3.1). For $n \in P_i$, write $m = Dn$. Then

$$a_{i,n} = cm = m + \eta_m, \qquad \eta_m = \frac{m}{q} > 0, \tag{6.6}$$

and, as $m \leq U$,

$$\eta_m \leq \frac{m}{2U^{13}} \leq \frac{1}{2} m^{-12}. \tag{6.7}$$

The choice of G_i changes none of F, D, H_i, P_i.

Proposition 6.1 (Finite extension)**.** *The enlarged configuration obtained by appending A_i is admissible. It still has $(D, H_j) = 1$ for every private prime currently present, and the new block satisfies*

$$W(A_i) > \frac{1}{24D}. \tag{6.8}$$

Proof. We first verify every type of ordered pair in the separation condition.

An old source and a new target. Fix $a = A_a/B_a \in F$, $n \in P_i$, and $k \in \mathbb{Z}$, and set $m = Dn$. Since $n \in \mathcal{R}(F, D)$, the number $ka - m$ is outside $(-1, 1]$. It belongs to $B_a^{-1}\mathbb{Z}$, so either

$$ka - m \leq -1 \qquad \text{or} \qquad ka - m \geq 1 + \frac{1}{B_a}.$$

In the first case $ka - a_{i,n} = ka - m - \eta_m < -1$. In the second, (6.5) gives

$$ka - a_{i,n} \geq 1 + \frac{1}{B_a} - \eta_m > 1.$$

Thus $|ka - a_{i,n}| > 1$ for every $k \in \mathbb{Z}$, in particular for every positive k. The denominator B_a is fixed for this source, and the argument places no upper bound on k.

A new source and an old target. By (6.4) and $c > 1$, every new point $a_{i,n}$ exceeds $M(F) + 1$. For $a \in F$ and $k \in \mathbb{N}$,

$$ka_{i,n} - a \geq a_{i,n} - a > 1.$$

Two distinct points in the new block. For distinct $n, n' \in P_i$ and $k \in \mathbb{N}$, primitivity implies $kn \neq n'$. Therefore

$$|ka_{i,n} - a_{i,n'}| = cD|kn - n'| \geq cD > D > 1.$$

Old pairs already satisfy (I1). The inequalities above also give the required ordering of the new block, so (I1) holds.

For (I2), consider first a preceding integer $N_{j,n}$. The old private primes do not divide it by induction, and the new prime does not divide it because $H_i > N_{j,n}$. Next consider a new integer $N_{i,n} = G_i Dn$. For every private prime H_j with $j \leq i$:

- $H_j \nmid D$, by the round's hypothesis for $j < i$ and by $H_i > D$ for $j = i$;
- $H_j \nmid n$, by $P_i \subset E$;
- $H_j \nmid G_i$, because G_i is a prime distinct from H_j.

Since H_j is prime, it does not divide $G_i Dn$. This proves (3.2) for every pair of indices now present. The size and distinctness requirements on the private primes follow from (6.1), and the reciprocal budget was preserved when H_i was selected. No assertion about an unchosen prime is used.

For (I3), equations (6.4) and (6.7) give

$$\frac{a_{i,n}}{H_i} = \frac{cDn}{H_i} > 4, \qquad 0 < a_{i,n} - Dn \leq \frac{1}{2}(Dn)^{-12}.$$

Finally, $n \geq 2D$ and $c < 2$ imply $cD \leq n$ and hence $\log(cDn) \leq 2 \log n$. Consequently,

$$\mathrm{w}(cDn) = \frac{1}{cDn \log(cDn)} \geq \frac{\mathrm{w}(n)}{4D}. \tag{6.9}$$

The map $n \mapsto cDn$ is injective. Summing (6.9) and using $W(P_i) > 1/6$ gives (6.8). □

7. Rounds and the infinite union

Proposition 7.1 (One round)**.** *Every admissible finite configuration can be extended, without changing any of its points or data, to an admissible finite configuration whose point set has gained weight at least* $1/576$*.*

Proof. Start with its point set F_0 and choose D by Lemma 4.1. Keep this D fixed, and prescribe $J = \lfloor D/12 \rfloor$ new blocks. After $s < J$ insertions, inequality (5.5) is valid. Sections 6 and 5 therefore allow the next insertion: the new configuration is admissible, its p_i satisfies $p_i \geq 2D$, and D remains coprime to every private prime. Finite induction performs all J insertions.

The blocks are pairwise disjoint by their ordering in (I1). Their total added weight is therefore, by (6.8), at least $J/(24D)$. Since $D \geq 48$, we have

$$J = \left\lfloor \frac{D}{12} \right\rfloor \geq \frac{D}{12} - 1 \geq \frac{D}{24}.$$

Thus the gain is at least $1/576$. The number of insertions is finite, even though no size bound on D is imposed. At the end of the round, (I2) allows Lemma 4.1 to choose a new grid using every integer $N_{i,n}$ now present. This new grid is used only for future blocks; no old value D_i is changed. □

Proof of Theorem 1.2. Begin with the empty admissible configuration and apply Proposition 7.1 repeatedly. Let F_R be the point set at the end of round R, and set $F_0 = \varnothing$. Each round extends the same preceding configuration. Thus

$$F_0 \subset F_1 \subset F_2 \subset \cdots, \qquad W(F_R) \geq \frac{R}{576} \quad (R \geq 1).$$

Set $A = \bigcup_{R \geq 1} F_R$. Every point is rational by (3.1), and every point exceeds 2, since $D_i \geq 48$ and $n \geq 2D_i$ at its insertion.

Given distinct $a, b \in A$, choose a round R with $a, b \in F_R$. The separation property of F_R yields $|ka - b| \geq 1$ for every $k \in \mathbb{N}$. This proves (1.1) on the union, since the round R containing the pair is chosen independently of k.

The set A is a countable union of finite sets. Its separation at $k = 1$ implies that every bounded interval contains only finitely many points, so A has no finite accumulation point. Equation (1.4) holds for every point by (I3), and remains true because no point is ever changed.

Finally, monotonicity of nonnegative sums gives

$$W(A) \geq W(F_R) \geq \frac{R}{576} \qquad \text{for every } R \geq 1.$$

Therefore $W(A) = \infty$. Since each point of A has finite weight, A must be infinite. This proves all assertions of the theorem. □

8. Order of choices and formal verification

8.1. **Quantifier order.** The dependency order for a new block is

$$\text{preceding configuration} \longrightarrow D \longrightarrow H_i \longrightarrow E \longrightarrow P_i \longrightarrow G_i \longrightarrow A_i.$$

The grid D is fixed throughout a round. Every use of Theorem 2.1 fixes the entire periodic set E, including all prime exclusions, before selecting P_i. Only then is G_i chosen. The analytic limit therefore has a fixed modulus. Compatibility with later primes is preserved by choosing each new H_i above all stored $N_{j,n}$, while the new labels and G_i avoid the current private primes.

8.2. **Choice of successive extensions.** The formal construction selects a witness of the proved finite-round extension theorem by classical choice and iterates this selection by recursion on the natural numbers. It establishes existence without a quantitative bound on the sizes of the grids, periods or blocks, or an extracted executable construction.

8.3. **Verification scope and reproducibility.** The complete existence theorem is formalized in Lean 4.24.0. All names below are in the namespace `Erdos143`. The two closed rational theorems are

`mainConclusion_proved`	`MainConclusion`
`strongConclusion_proved`	`StrongConclusion`

The first gives a countable rational set in $[2, \infty)$, separation for every ordered distinct pair and every positive natural multiplier, and infinite nonnegative total weight. The second adds local finiteness and (1.4). Their definitions are in `Erdos143/Definitions.lean` and their proofs in `Erdos143/Main.lean`. The project proves the periodic selection lemma and the finite-round extension theorem, then constructs the nested sequence using classical choice as described above.

The real-domain statement is proved in `RealConclusion.lean` as

`Erdos143.realConclusion_proved`.

It explicitly includes countability, rationality, the lower bound 2, every ordered multiplier inequality, and both divergence of the nonnegative sum and nonsummability of the real-valued weight. The transfer uses the injective embedding of $\mathbb{Q}$ into $\mathbb{R}$ and an equivalence of the indexing sets. In particular, divergence is encoded by $\top$ in the extended nonnegative reals (Lean's `ENNReal`), together with an explicit negation of `Summable`; it is not inferred from the conventional value of a nonsummable real `tsum`. The lower bound and divergence imply that the set is infinite.

The periodic selection theorem is checked by the declaration

`periodicPrimitiveSelection_proved`.

The finite construction follows the same order of choices and uses the same strict prime budget, closed residue filter and product reset grid as the proof above. The declarations

`insert_block_exists`	finite insertion
`block_extension_valid`	simultaneous invariants
`round_extension_exists`	gain of at least $1/576$
`stateSequence_mass_lower`	prefix weight bound

connect these steps to the nested union in `Main.lean`. The formal states store finite sets of triples (H, N, q) and private primes; the prose groups the triples into blocks. A detailed statement-by-statement correspondence is supplied with the code in the `docs/` directory. No asymptotic assertion for every individual Ω level is used. The supplementary file `ManuscriptAudit.lean` also checks a single existential statement combining these conclusions, including infinitude and the same nested finite prefixes.

The environment pins mathlib to revision

`f897ebcf72cd16f89ab4577d0c826cd14afaafc7`,

with all transitive revisions fixed in `lake-manifest.json`.

From the supplied supplementary project, the commands

```
bash scripts/bootstrap.sh
bash scripts/verify.sh
```

retrieve the pinned dependencies, build the proof with warnings treated as errors, audit all proved declarations and compile the expanded statement checks. Network access and the tools listed in the project `README.md` are required. On platforms other than Linux x86_64, Lean 4.24.0 must first be installed using the platform's standard elan installation.

The supplementary material contains the two original local verification records and a further build and statement audit dated October 23, 2026. The original runs include a separate-copy clean rebuild of the project using pinned dependency caches. These checks do not constitute a from-source rebuild of Lean and mathlib. All 245 proved declarations passed the axiom audit. The three closed conclusion theorems use only `propext`, `Classical.choice` and `Quot.sound`, the standard foundations of the Lean/mathlib formalization; no added analytic or arithmetic axiom is used. The original frozen source archive, its SHA-256 manifest and the local logs are included in the supplementary project's `verification/` directory. Typesetting this article does not execute the formal proof. Kernel checking establishes the stated formal results from their recorded foundations. Independent expert assessment of the formulation, its correspondence to Erdős's question and the exposition remains part of the journal's review process.

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